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Feature Focused Scene Classification

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Abstract

To accomplish the projects goal of outdoor scene classification, the chosen approach focused on differentiating features for optimal category separation. The constraints of limited resolution and abridged training set were not considered in feature selection. Instead, features were chosen based on heuristic observations made during the compulsory resolution experiment. The variation in resolutions revealed the first feature, frequency, as a major component of class identity. Color composition was identified as a possible feature after observing its consistency across low resolution test images. The Scale Invariant Feature Transform (SIFT) was used for the last major feature because of its history as a discriminating feature among complex scenes. The system design was then based around finding a configuration of these features to produce accurate scene classifications. To perform the final classification, two options were evaluated; a single layer consisting of dimensionality reduction and a support vector machine and multiple layers, each with their own support vector machine. These two options were chosen based on the variability of the features and relative independence of the classifier and feature selection. As a group of three, the goal was to provide three strongly discriminating features that would be combined to form a system more capable than any sole feature. The results of the independent frequency response gave clear separation between classes having high variations in the spatio-frequency domain. Color histograms provided high recognition rates for forest, highway and building classes. The SIFT method also separated forest, highway and building classes accurately. Gabor filters served well by themselves but not when in conjunction with other classes.

1 Introduction

Scene recognition in image processing not only consists of recognizing the objects, but also their holistic composition that forms the scene. While detection of individual objects helps in classifying a scene to some degree, the relationship between the objects is major deciding factor in scene classification. Identification of man-made objects, and describing the scene according to their functional representation is an easier task than labeling a natural scene, which consists of multiple, ambiguous object classes and varying object to object relationships. This study takes an aim multiple feature sets produced from natural and artificial feature classification methods. To obtain such classification, we use Gabor Features for object detection and feature space separation. Gabor features, described by Gabor filters

2 Methods

Each feature was investigated independently before being combined with other features. Features were evaluated using statistical methods as well as individual classification results. Statistical evidence was gathered primarily from the training set as inclusive or exclusive validation runs. This would provide further insight into how the feature would perform against similar images or how well the feature generalizes its training set. Classification results were generated from a exclusive subset of the training set as well as the test set.

The Gabor feature extraction consists of (1) designing a set of Gabor filters in the spatio-frequency domain[4], and (2) forming a bank of filters to be applied to the images for feature matching [2]. While step (1) makes sure that optimal filters are designed to match feature variability of natural and man made objects, in step (2) we decide the ideal filter parameters in a set of Gabor filters and form a filter bank. The goal is to produce feature vectors that describe the scene as accurately as possible.

2.1 Gabor Filters

Gabor filters have long been used for texture detection, segmentation and texture synthesis. The reason for their popularity is because they seem to mimic the characteristics of cells in the visual cortex of macaca and cats [3] [1]. Appropriate Gabor filters can approximate these characteristics. Mathematically, a Gabor filter of a particular orientation and frequency is a sinusoid modulated by a Gaussian:

$$h(x, y) = s(x, y)g(x, y)$$

Where $s(x, y)$ is a sinusoidal plane and $g(x, y)$ is a 2-D Gausssian. The Gaussian is represented by:

$$g(x, y) = \frac{1}{\sqrt{2\pi\sigma}} e^{-1/2(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2})}$$

and $s(x, y)$ as:

$$s(x, y) = e^{-j2\pi(u_0x + v_0y)}$$

A Gabor filter in 2 dimensions can thus be used to approximate the frequency and spatial response of the different classes.

2.1.1 Gabor Filter Bank

A filter bank of 3 orientation and 8 frequencies was considered for feature extraction, giving a total of 24 filters for feature vector creation. Images from each class in the test set was convolved with the filter bank to give a 256x1 feature vector. The criterion of selecting particular frequencies was mostly based on the natural dissimilarity and the apparent inter-class separation. For example, features belonging to a set of natural images, like coast, appears considerably different than a set of artificial images, such as tall buildings. It was hypothesized that the feature set resulting from the filtered images would obtain a greater inter-class separation, while relying on other methods (explained) below for local feature extraction and intra-class separation. As the following table shows, the intra-class separation is not so prevalent with just the use of Gabor filter bank, while good separation is achieved between the classes. It should be mentioned that classes such a forest, which depict a uniform frequency distribution, are the ones best separated from other classes. On the other hand, coast does not perform well owing to sparse frequency distribution and many low frequency, blended features:

Intra-class Separation	
Class	Value
COAST	3.8339
HIGHWAY	7.2859
FOREST	5.1482
TALL BUILDING	7.9352
MOUNTAIN	5.4442

The above separation metric is defined as [8]:

$$J(w) = \frac{|m_1 - m_2|^2}{s_1^2 + s_2^2}$$

Where m_1, m_2 and s_1, s_2 are the means and standard deviations of the two classes, respectively. The goal is to maximise the value of $J(w)$. The table below shows the Inter-class distance from the feature vectors obtained by the Gabor filter bank:

Inter-class Dist.	
Class, Class	Value
COAST, HIGHWAY	3.8339
HIGHWAY, FOREST	7.2859
FOREST, TALL BUILDING	5.1482
TALL BUILDING, MOUNTAIN	7.9352
MOUNTAIN, COAST	5.4442

To understand how feature vectors differ from each other, not in just terms of intensity values, but the overall features, Pearsons' correlation coefficient, r , defined as below was used:

$$r = \frac{\sum_i (x_i - x_m)(y_i - y_m)}{\sqrt{\sum_i (x_i - x_m)^2} \sqrt{\sum_i (y_i - y_m)^2}}$$

Where x_i is the intensity of the i th pixel in the image 2, y_i is the intensity of the i th pixel in image 2, x_m is the mean intensity of image 1, and y_m is the mean intensity of image 2. The result from the inter and intra class correlation is shown below:

Inter-class Correlation					
	Mountain	Coast	Highway	Tall Building	Forest
Mountain	0.4953				
Coast	-0.1776	0.5856			
Highway	-0.4508	0.3020	0.8710		
Tall Building	-0.0852	-0.3628	-0.1010	0.2680	
Forest	-0.5244	-0.1160	0.0048	-0.2992	0.9880

2.1.2 Gabor Energy

Gabor Energy features are just the absolute sum of the squares of the real and imaginary part. The feature vectors from the real and imaginary parts of the filters were combined to explore features not otherwise apparent with the Gabor filter bank. This is also known as motion energy, which is equal to the pair summation response with a phase difference of $\pi/2$. This quantity is defined as follows:

$$E_{v,\theta}(x,y,t) = \sqrt{r_{v,\theta,0}^2(x,y,t) + r_{v,\theta,\pi/2}^2(x,y,t)}$$

Where summations from $\theta = 0$ and $\theta = \pi/2$ give the real and imaginary responses. The Intra and Inter class separation obtained from the above method, using the separation metrics and correlation coefficient is given below:

Intra-class Separation	
Class	Value
COAST	1.0862
HIGHWAY	4.0757
FOREST	1.8493
TALL BUILDING	1.7036
MOUNTAIN	1.0292

Inter-class Dist.	
Class, Class	Value
COAST, HIGHWAY	9.7360
HIGHWAY, FOREST	1.4306
FOREST, TALL BUILDING	7.0072
TALL BUILDING, MOUNTAIN	2.0812
MOUNTAIN, COAST	2.8633
MOUNTAIN, HIGHWAY	9.7076

2.1.3 Power Spectrum Density

Power spectrum provides the higher-order statistics of natural images, and helps represent the neuron activity in the visual system and our adaptation to it. To measure the power spectrum density, a simple Fourier Transformation to convert an image from space to frequency domain of each testing image was calculated, and was then shifted to a zero frequency in the middle. Square of the modulus of the Fourier transformation gives us the power spectrum density.

The power spectrum of each image was contrast-normalized, to give a measure of total neighbourhood activity in its region. This was achieved by convoluting a power spectrum image with a Gaussian (for a Gaussian Neighbourhood representation) and dividing by its square. The result achieved was the following:

Inter-class Correlation					
	Mountain	Coast	Highway	Tall Building	Forest
Mountain	0.3465				
Coast	-0.1308	0.8904			
Highway	-0.3684	0.0345	0.9359		
Tall Building	-0.0299	-0.3628	-0.2321	0.6684	
Forest	-0.3314	-0.1160	-0.5097	-0.5550	0.9300

2.1.4 Template matching

Template matching is a Computer Vision technique that segments the image according to a pre-defined response value, which is a template derived from either the target image itself or a set of images. This template is then applied to the target image to figure out the relation between the two. This relation is more of the similarity or dissimilarity in behaviour of the two images rather than the difference in the value of pixels. The output image is then mapped over to the original image to locate the relationship measure, which in this case are tree crowns. This technique is mathematically known as cross-correlation. Gray-level cross correlation is defined by: correlation computes a numerical index which indicates how accurately or behaviorally the template matches the image in that position on a pixel by pixel basis. Mathematically, correlation is the degree to which two variables agree by behaviour. Correlation for gray level images can be computed using the following equation:

$$corr(x, y) = \frac{\sum_i^N [(x_i - \bar{x}) (y_i - \bar{y})]}{\sqrt{\sum_i^N [(x_i - \bar{x})^2 (y_i - \bar{y})^2]}}$$

where x is the template gray level image, $\bar{x}$ is the average gray level in the template image, y is the source image currently under being processed, $\bar{y}$ is the average gray level in the source image, N is the number of pixels in the section image. The value of correlation is always between -1 & $+1$, larger the value stronger the relationship between template and source patch and vice versa. Here, correlation is applied between the average templates for a total of 6 images from each class and the source image, revealing the relationship between the two.

Correlation Coefficient					
	Run 1	Run 2	Run 3	Run 4	Run 5
Mountain	-0.4990	0.2246	-0.1363	0.4730	0.1236
Coast	0.0693	-0.2903	-0.2495	0.2655	0.6412
Highway	-0.0575	-0.0300	0.5045	0.0783	0.2316
Tall Building	0.1726	0.0790	0.4144	0.3740	-0.0959
Forest	0.3789	0.2045	-0.0331	0.1637	0.0414

3 Color Histograms

As a perceptual feature, color composition had a clear impact on the separation between the chosen classes of outdoor scenes during the resolution experiment. As a method for scene classification, color information is historically one of the oldest machine recognized features [6]. The simplest method presented for scene categorization used a non-linear mapping of the RGB color space and histogram intersection [7]; this provided adequate separation between indoor and outdoor scenes.

To classify different types of outdoor scenes, a similar approach was considered. Each of the training images was separated into 16 equally sized blocks, a clear advantage of the single square resolution of the image set. Once the blocks were separated, the inner four and bottom two blocks were averaged across each class to create a template. The template would then contain both color and limited spacial information. The selection of the middle and bottom blocks allowed any noise on the edge of the image or any non-discriminating features such as the sky to be removed from the average. The average block histograms were then used to compute the histogram intersection with each test image. The resulting histogram intersection, still partitioned into blocks, along with the test images histogram was concatenated and passed into a support vector machine for classification.

3.1 Results

The results of the color histogram test are organized by color space. The results listed below are the number of mis-classifications (false positives & false negatives) over 150 test images with 30 test images per class. The number of principle components as well as the SVM kernels are listed as parameters.

Color Space Comparison	Coast	Forest	Highway	Mountain	Building
RGB	linear	linear	linear	quadratic	linear
PC = 55	36	4	22	35	31
YCrCb	quadratic	linear	linear	linear	linear
PC = 55	27	1	11	19	14
HSV	quadratic	quadratic	quadratic	quadratic	quadratic
PC = 55	25	4	9	29	28
LAB	linear	linear	linear	linear	linear
PC = 50	40	1	4	56	5

The best performing color space was the LAB and YCrCb, with the best performing classes being forests, highways, building.

The table below highlights the percentage error for the best case of each class.

Least % Error	Coast	Forest	Higway	Mountain	Building
RGB	18	16	12	18.66667	19.33333
YCrCb	18	0.666667	7.333333	12.66667	9.333333
HSV	16.66667	2.666667	6	19.33333	18.66667
LAB	16	0.666667	2.666667	20.66667	3.333333

The table below lists the false positives, which heavily influences the choice of classifiers.

False Positives	Coast	Forest	Highway	Mountain	Building
RGB	0	4	3	6	12
YCrCb	10	0	5	11	7
HSV	2	2	1	8	17
LAB	6	0	4	11	1

LAB has fewer false positives than YCrCb, which combined with the high accuracy makes LAB the best candidate for classification.

The table below lists the results of the LAB color space tests. The main diagonal lists the false negatives while the off-diagonal entries list the false-positives.

LAB	Coast	Forest	Highway	Mountain	Building
COAST	18	0	3	6	1
Forest	0	1	1	0	0
Highway	2	0	0	0	0
Mountain	3	0	0	20	0
Building	1	0	0	5	4
Total % Error	16	0.666667	2.666667	20.66667	3.333333

The combination of other features will further improve the final classification.

4 SIFT Features

SIFT-Based features were investigated and implemented as another source of image features for separation between the classes. There are two main reasons SIFT was chosen- firstly, it is well established and has been demonstrated to result in meaningful, distinct data. Secondly, the multiple kinds of data retrieved from SIFT (both the patch descriptors and the location data) gives multiple avenues to explore for features.

4.1 Bag of Words

The first method investigated was a simple bag of words approach. This is a well-tested method that has performed well in the past, and fairly simple to implement. Firstly, SIFT Feature Descriptors are generated for each of the images in the entire training set, with no separation between classes. These are clustered to N different groups using k-means clustering. The end result of this is the Dictionary of words that will be used to generate and count the bag of words.

The feature vector generated from an individual image is a count of the matched words between the SIFT points on the image and the clustered points in the dictionary. Each of the SIFT descriptors found in a given image is matched to one of the cluster centers by simple 1-nearest neighbor. The index of the matching cluster is then taken, and the corresponding index in the bag is incremented. The end result is taken as the feature vector for later combination with the other feature sets.

The difficulty with this approach was to decide the number of clusters (or words) to create during the k-means clustering. As such, dictionaries were generated with 16, 32, 64, 128, and 256 clusters. With these dictionaries, the feature vectors were generated for each of the testing images. Once the feature vectors were computed, the average distance between each vector, and every vector of another class, was computed. This is defined as:

$$D(A, B) = \frac{\sum_{j=1}^m \sum_{i=1}^n |F_{A,i} - F_{B,j}|}{ij}$$

Where $F_{A,i}$, is the i th feature vector of an arbitrary class A, $F_{B,j}$, is the j th feature vector of an arbitrary class B (which may be identical to class A), n is the cardinality of class A, and m is the cardinality of class B. This gives a rough sense of the separation between the two different classes (or, in the case of A=B, the spread of the class).

16 Clusters					
	Coast	Forest	Highway	Mountain	Building
COAST	41.5315	79.2108	46.025	48.7852	58.2162
Forest	79.2108	47.0654	79.6979	60.0117	67.441
Highway	46.025	79.6979	49.6027	51.0014	61.4305
Mountain	48.7852	60.0017	51.0014	39.9711	51.6121
Building	58.2162	67.441	61.4305	51.6121	57.3855

32 Clusters					
	Coast	Forest	Highway	Mountain	Building
COAST	37.8374	64.2169	41.9903	42.7518	51.5165
Forest	64.2169	40.0487	65.1388	49.3497	58.0957
Highway	41.9903	61.1388	43.433	44.8522	54.7702
Mountain	42.7518	49.3497	44.8522	35.1846	46.6972
Building	58.2162	67.441	61.4305	51.6121	57.3855

64 Clusters					
	Coast	Forest	Highway	Mountain	Building
COAST	33.4241	49.9647	36.6037	36.6691	46.1834
Forest	49.9647	34.9264	50.9914	40.4617	50.1090
Highway	36.6037	50.9914	37.4752	238.3067	48.5068
Mountain	36.6691	40.4617	38.3067	31.4833	43.1413
Building	46.1834	50.1090	48.5068	43.1413	50.5142

128 Clusters					
	Coast	Forest	Highway	Mountain	Building
COAST	31.2692	41.1383	33.6404	33.0483	46.1834
Forest	41.1383	30.7087	42.1109	34.466	50.1090
Highway	33.6404	42.1109	34.0766	34.2223	48.5068
Mountain	33.0483	34.466	34.2223	29.1172	43.1413
Building	42.2687	43.986	43.9735	39.9594	50.5142

256 Clusters					
	Coast	Forest	Highway	Mountain	Building
COAST	29.7933	35.0256	31.0659	29.8077	38.1696
Forest	35.0256	28.1204	36.3822	30.405	38.9146
Highway	31.0659	36.3822	31.6752	31.2106	39.9356
Mountain	29.8077	30.405	31.2106	26.7783	26.416
Building	38.1696	38.9146	39.9356	36.416	42.8012

From the gathered data, it was decided that 64 clusters would be optimal for the separation between classes- while the distances were not the absolute highest, the inter-class distances were on average higher in relation to the intra-class distances (the class spreads), indicating that 64 clusters gave the best separation between classes. The class separation was also considered to be sufficient for this method to be used in the classifier. Specifically, most classes were found to have good separation between themselves and the other classes, with the exception of tall buildings- this class, for all cluster sizes, generally had higher spread than distance to other classes.

An additional modifier to this method was also examined. The feature vectors were normalized before the distances were calculated, in an effort to examine the distances between the general patterns of the feature vectors rather than their absolute values. However, this resulted in worse separation across all examined cluster counts, so the method was discarded.

As a precursor to the final classifications using all feature vectors, the Bag of Words was evaluated as a single feature vector for classification. An SVM was trained for each class, using the 70 training images in that class as positive training data, and all other training images as negative training data. The following confusion matrix is the raw output from the SVMs, including classifications that returned multiple positive results:

	Coast	Forest	Highway	Mountain	Building
COAST	13	0	2	11	0
Forest	0	28	1	1	0
Highway	3	0	29	2	0
Mountain	7	1	1	11	0
Building	1	0	0	7	24

In general, the classes had strong responses along the diagonal. However, the results were much less promising when removing any classifications with more than one positive result:

	Coast	Forest	Highway	Mountain	Building
Coast	13	0	2	11	0
Forest	0	28	1	1	0
Highway	3	0	29	2	0
Mountain	7	1	1	11	0
Building	1	0	0	7	24

However, due to the nature of the final classifier (combining information from multiple sources of features), this non-specificity is not as large an issue.

4.2 SIFT Density Feature

The other SIFT-based feature that was examined was a method of determining a profile of the density of SIFT feature points that occurred in an image. It was hoped that this would result in separation between busier scenes (tall buildings, forest) and smoother, less busy scenes (highways, coasts).

The method is as follows: After the SIFT features are computed, the image is split into a number of bins, $N \times N$ on each side. The number of SIFT feature points that fall within each bin are counted, and entered into a histogram. This histogram is the final feature vector. The size of this binning was tested at 4×4 , 8×8 , 16×16 and 32×32 . However, all of the results looked similar to the following:

	Coast	Forest	Highway	Mountain	Building
COAST	13	0	2	11	0
Forest	0	28	1	1	0
Highway	3	0	29	2	0
Mountain	7	1	1	11	0
Building	1	0	0	7	24

Little meaningful separation between the classes was extracted using this method. In some cases, the classes were farther apart from themselves than from other classes. In the cases where this was not true, the class separations are not generally significantly more distant than the intra-class distance. As such, this method was not deemed useful for the final classifier.

5 Compiled Feature Vectors and Classifiers

Each of the methods of feature extraction gives rise to good separation between several classes, and each method has classes that do not separate well. This is as was expected, and was the motivation behind this project, so the task now is to combine them into the same classification system. Two separate methods to do so were investigated. The first, referred to as the Concatenation Method, involves concatenating together the different feature vectors, and then classifying using SVMs. The other method, referred to as the Hierarchical Method, involves using multiple different levels of SVM classifiers in a tiered structure.

5.1 Concatenation Method

The concatenation method is the simpler of the two methods investigated. The feature vectors from each of the feature extraction methods for a given image are concatenated into one longer vector, in an arbitrary order. Principal Component Analysis is employed on these Compiled Vectors, with two aims in mind- one, to reduce the dimensionality of the feature vectors for computational ease; and two, to extract the most meaningful dimensions of the compiled features. First, both the SIFT bag of words and the color histogram features were concatenated.

	Coast	Forest	Highway	Mountain	Building
Coast	13	0	2	11	0
Forest	0	28	1	1	0
Highway	3	0	29	2	0
Mountain	7	1	1	11	0
Building	1	0	0	7	24

This shows improvement over the SIFT features in all classes- positive results along the diagonal are higher in all cases. Compared to the LAB color space histogram results, Coast shows moderate improvement and Mountain shows significant improvement, while the other classes show a bit of degradation.

The concatenation method was also examined with the addition of the Gabor filter features.

	Coast	Forest	Highway	Mountain	Building
Coast	14	0	28	2	3
Forest	4	26	3	3	5
Highway	5	0	26	1	3
Mountain	2	3	9	18	5
Building	1	0	13	6	16

Between the two-vector concatenation, and the three-vector concatenation, in general the two-vector concatenation performed better. The tall building class performed significantly worse in the three-vector, and the forest and highway classes performed very slightly worse. Overall, adding the gabor filter feature vectors did not seem to improve the results.

5.2 Hierarchical Method

The hierarchical method takes advantage of the fact that classifying using SVMs can give more than one positive result for any given image. Firstly, the feature extraction methods were sorted in order of their broadness, defined as the average number of positive results returned for an image when classified using only that feature vector.

Broadness	Value
SIFT Bag of Words	1.3800
Gabor Filters	1.3067
Color Histograms	0.8600

In this method, the images are first classified by the broadest classifier (Bag of Words). If this results in exactly one positive result, this single result is taken as the classification. If there are no results, classification is considered to have failed. If there are more than one result, the image is then classified by the next most narrow classifier (gabor filters), but is restricted to only the classes that were positive in the previous result. The process is repeated, and the results are treated the same. If an image is tested on the final method (Color histograms), and the result is not a single positive output, classification is considered to have failed.

Again, this method was first tested with only the SIFT features and the color histogram features:

	Coast	Forest	Highway	Mountain	Building
Coast	7	0	7	1	0
Forest	0	27	1	1	0
Highway	2	0	27	1	0
Mountain	1	1	4	8	4
Building	1	0	2	3	21

Results are fairly similar in pattern to the Concatenation method, but are worse in every category, though not to a significant degree.

Next, the Gabor Filters were used as an intermediate step between SIFT Bag of Words and Color Histograms.

	Coast	Forest	Highway	Mountain	Building
Coast	8	0	3	2	2
Forest	0	23	0	1	2
Highway	4	0	14	0	1
Mountain	3	1	2	9	4
Building	2	0	1	4	6

Again, as with the concatenation method, Gabor filters seemed to worsen the performance except for negligible gains in coast and mountain.

The table below shows the average of the concatenation and hierarchical methods, with the Gabor feature vectors.

Average Accuracy - Concatenation					
	Coast	Forest	Highway	Mountain	Building
% Correct	44.3333	95	34.3333	48.3333	66.6666

The next table shows the average accuracy of the Hierarchical Method, with the SIFT, Color and Gabor feature combined together:

Average Accuracy - Hierarchy					
	Coast	Forest	Highway	Mountain	Building
% Correct	25	83.3333	68.3333	28.3333	45

Here is the total average accuracy of the Hierarchical and Concatenation methods:

Average Accuracy - Hierarchy + Concat					
	Coast	Forest	Highway	Mountain	Building
% Correct	34.5	89.16665	51.3333	38.3333	55.8333

6 Conclusion

As evidenced by the results of the combined and hierarchical classifier, the features selected had overlapping performance characteristics. For classes like forest, highway, and building, all classifiers performed relatively well (90%+ accuracy); but for the problematic classes of coast and mountain, no feature alone or combined produced noticeably better classification decisions. This result shows that different features can arrive at the same classification accuracy and that unique feature selection alone cannot ensure that each class will be adequately separated.

The problem lies in the visual structure of the class, where no feature could extract a unique response for nondescript or highly variable classes. This is clearly the problem with coast and mountain, where the visual features alone are so varied that without semantic knowledge, many scenes in the same class would be considered very different. The color of the coasts varies greatly depending on how much of the beach is visible and the variation of the water and sky colors.

The mountain class included calm snow covered mountains as well as mountains in forests and near rivers, which would confuse any collection of interest points with the wide range of settings. Frequency was again fooled by the noisiness of the scene, where noisy scenes could be part of any particular class and could not be separated.

The system performed scene classification on three of the five classes exceptionally well for using a straightforward approach for feature selection and classification, and was able to outperform random chance on the remaining two classes. For a future improvement, some semantic information could be included to help classify coast and mountain scenes based on low level cues from the features explored in this system.

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